

Signing a graph to have small magnitude eigenvalues

OPG-48264

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Conjecture If A is the adjacency matrix of a d -regular graph, then there is a symmetric signing of A (i.e. replace some $+1$ entries by -1) so that the resulting matrix has all eigenvalues of magnitude at most $2\sqrt{d-1}$.